

Agenda

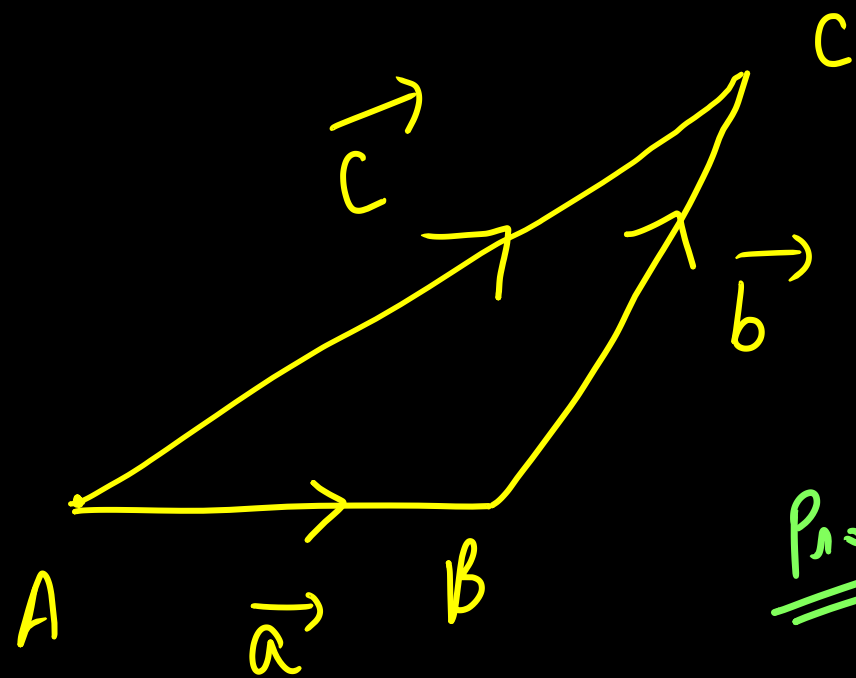
→ Triangle law of vector

→ w is $\perp$ to $w^T x + w_0 = 0$

→ Perpendicular dist. of a point x_0
from $w^T x + w_0 = 0$

→ Half space

Triangle law of vector addition



$$\vec{AB} + \vec{BC} = \vec{AC}$$

Proof

$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$$

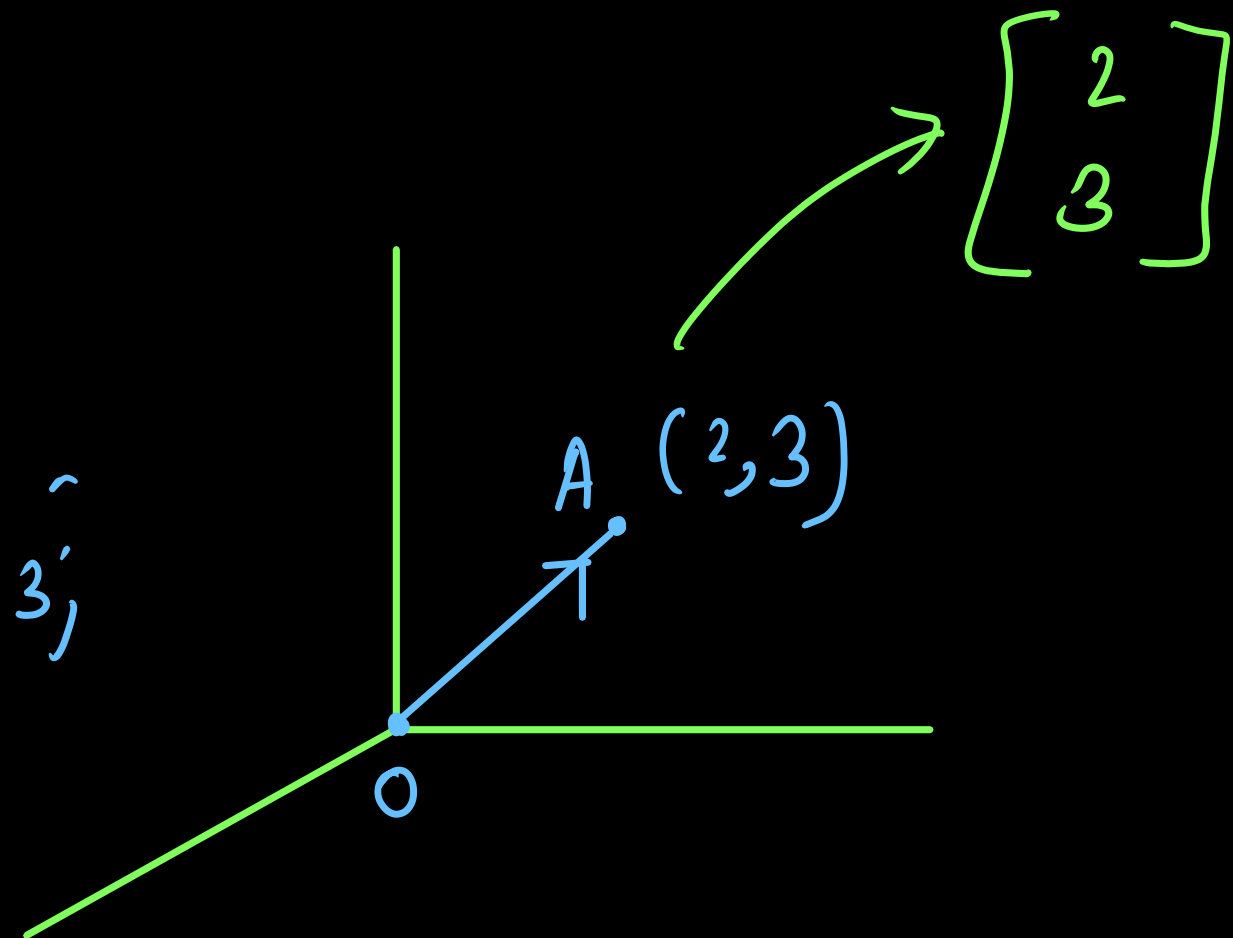
$$\begin{aligned}\vec{AB} + \vec{BC} &= -\vec{CA} \\ &= \vec{AC}\end{aligned}$$

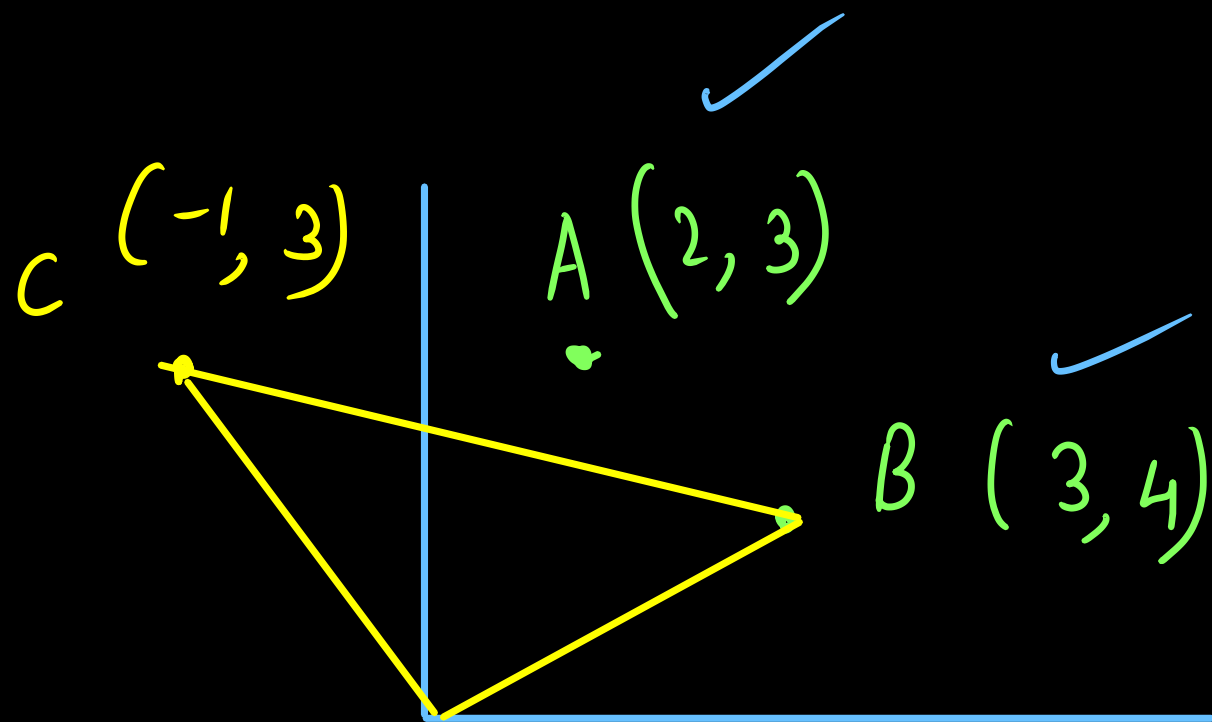
Position Vector ✓✓

A point is represented by a vector from origin to that point.

A (2, 3)

✓ $\vec{OA} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2\hat{i} + 3\hat{j}$





→ What is $\vec{AB}$?

$$\vec{AB} = \vec{OB} - \vec{OA}$$

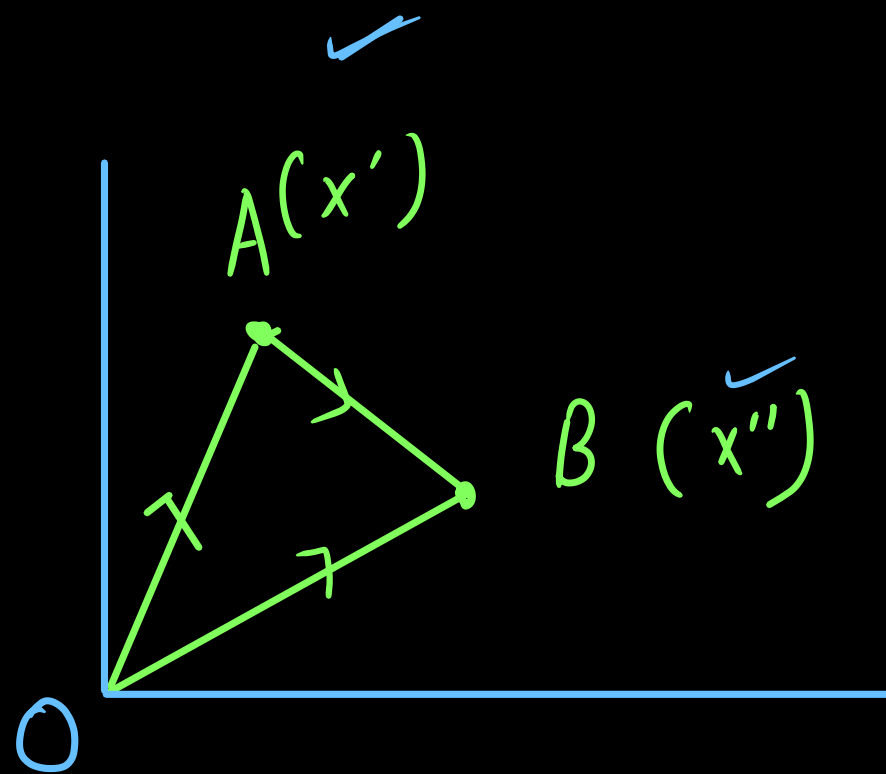
$$\vec{BC} = \vec{OC} - \vec{OB} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\vec{OB} = \vec{OC} + \vec{CB}$$

$$\vec{OC} - \vec{CB} = \vec{OB}$$

$$= \begin{bmatrix} -4 \\ -1 \end{bmatrix}$$

$$= \hat{i} + \hat{j} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

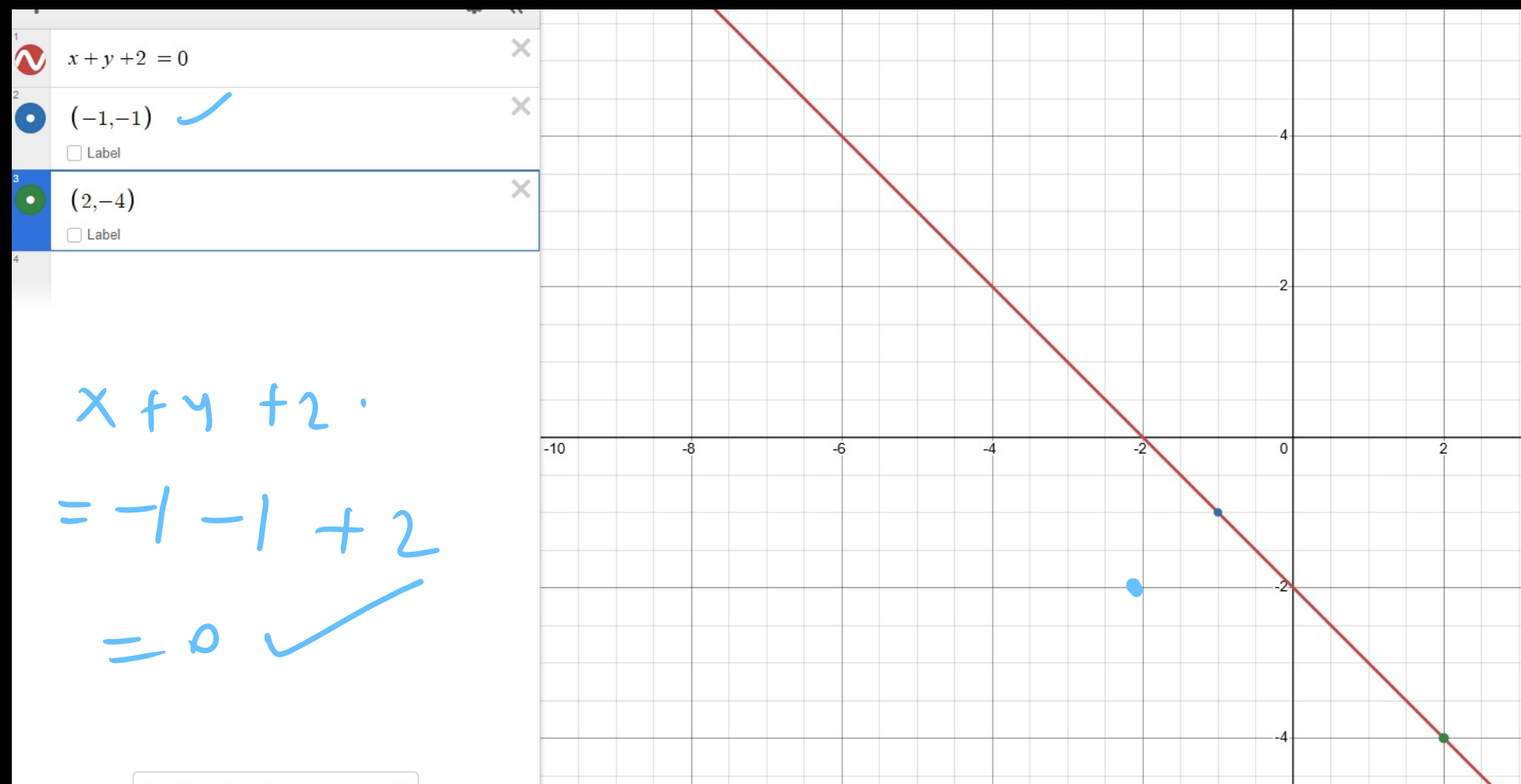


→ What is $\vec{AB}$?

$$\vec{OA} + \vec{AB} = \vec{OB}$$

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= \vec{x''} - \vec{x'}\end{aligned}$$

{ $\rightarrow$ If a point x' lies on line $w^T x + w_0 = 0$
 $w^T x' + w_0 = 0$

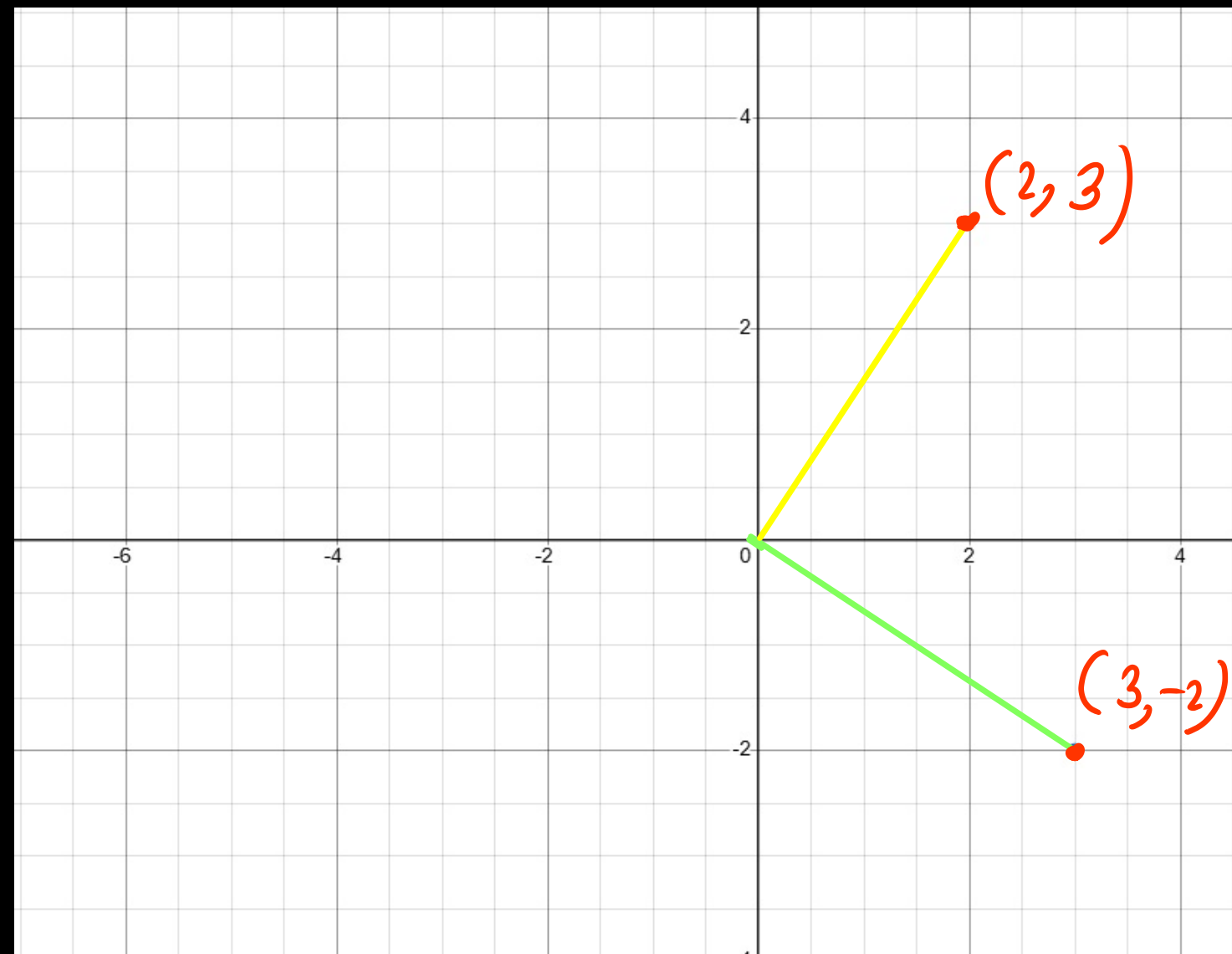


$(2, 5)$
 $\downarrow$
 $x + y + 2 = 9$ ✓
 $-2 - 2 + 2 = -2 < 0$ ✓

•> Consider 2 vectors $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $y = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

What is the dot product? $\rightarrow 2 \times 3 - 3 \times 2$
 $= 0$

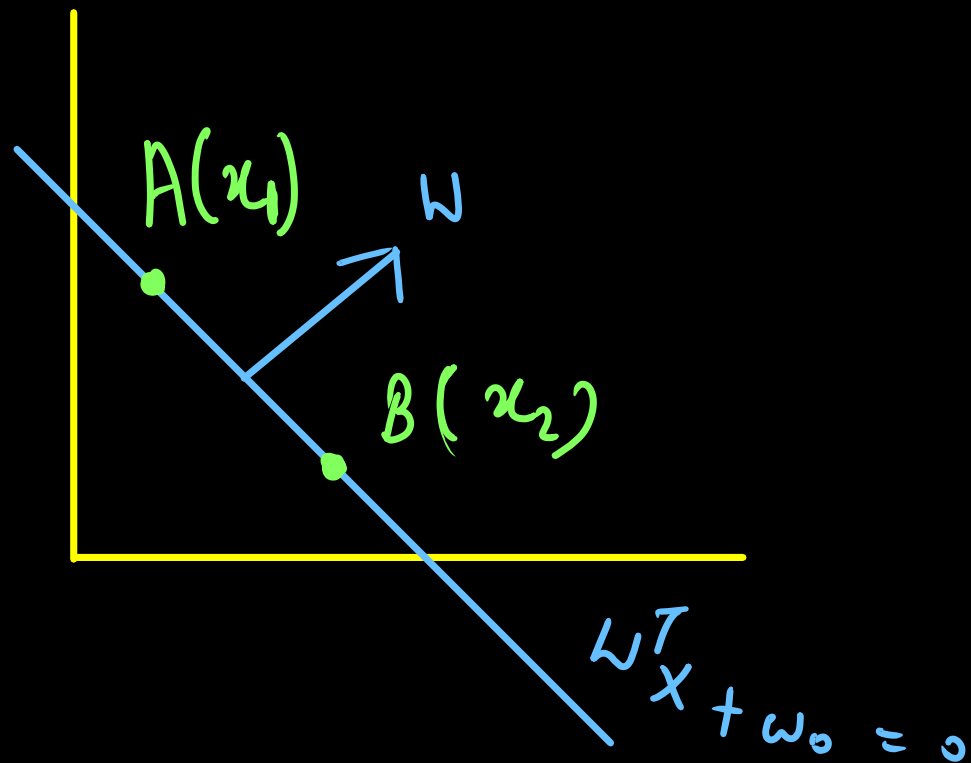
What is the angle b/w them? $\rightarrow \cos^{-1}(0)$
 $= 90^\circ$



→ Prove that

$$\vec{w}^T \vec{x} + w_0 = 0$$

$\vec{w} \perp$ to line/hyperplane
(Normal)



$$\vec{w}^T \vec{x}_1 + w_0 = 0 \quad (1)$$

$$\vec{w}^T \vec{x}_2 + w_0 = 0 \quad (2)$$

Subtract (1) from (2)

$$\vec{w}^T (\vec{x}_2 - \vec{x}_1) = 0$$

$$\vec{w}^T \vec{AB} = 0$$

$$\vec{w} \perp \vec{AB}$$

$$\vec{w} \perp \vec{w}^T \vec{x} + w_0 = 0$$

1. Given the vectors $a = 3\hat{i} + 4\hat{j}$ and $b = \hat{i} + \hat{j}$, what is the scalar projection of a onto b ?

A. $\frac{7}{5}$

B. $\frac{1}{\sqrt{2}}$

C. $\frac{7}{\sqrt{2}}$

D. 7

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\frac{a \cdot b}{\|b\|} = \frac{3 \times 1 + 4 \times 1}{\sqrt{1^2 + 1^2}}$$

$$= \frac{7}{\sqrt{2}}$$

1. Given the vectors $a = 3\hat{i} + 4\hat{j}$ and $b = \hat{i} + \hat{j}$, what is the scalar projection of a onto b ?

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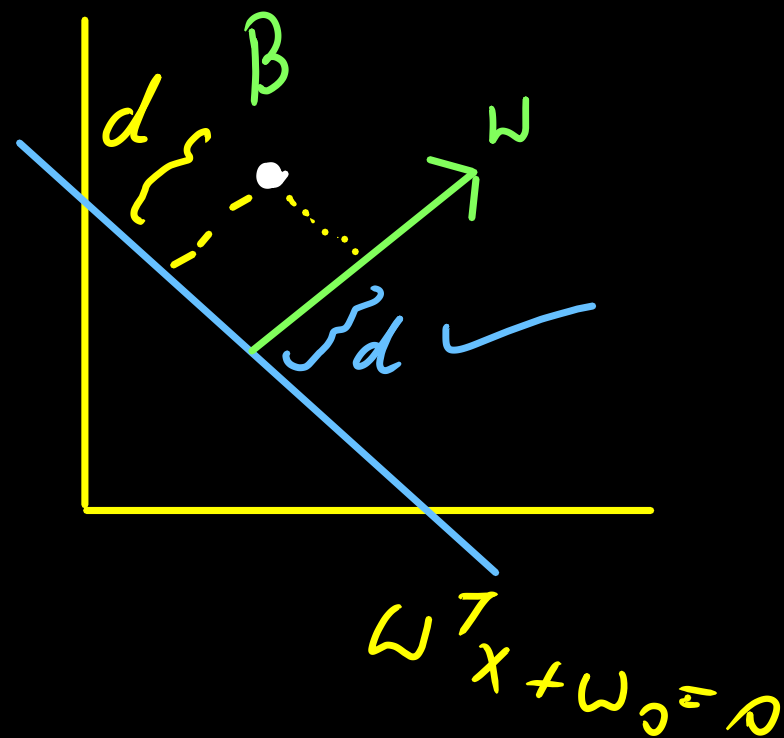
✓ That's right!

Applying the formula for the scalar projection, $\frac{a \cdot b}{|b|}$, gives

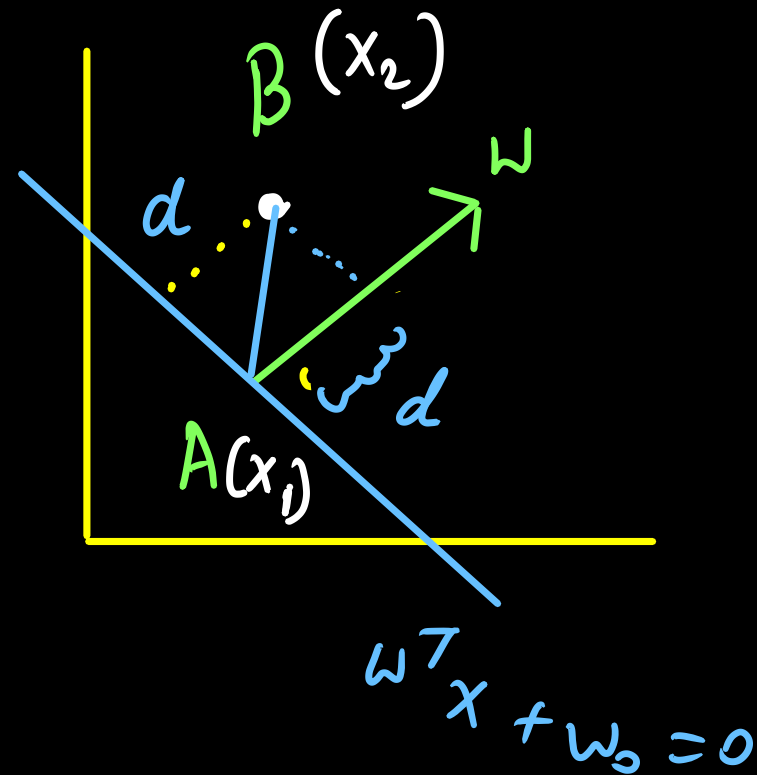
$$\frac{(3)(1) + (4)(1)}{\sqrt{1^2 + 1^2}} = \frac{7}{\sqrt{2}}.$$

D. 7

→ Perpendicular distance b/w a point & plane.



→ Perpendicular distance b/w a point & plane.



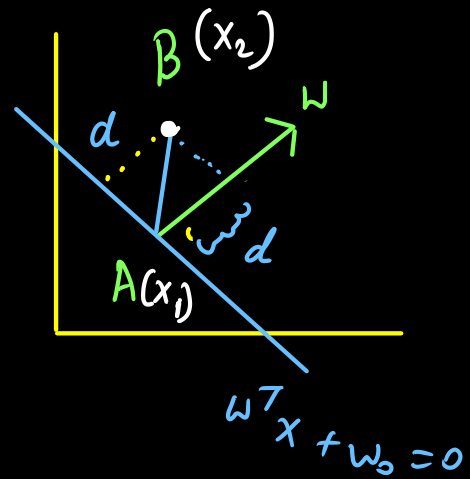
→ How can I calculate d ?

✓ $\vec{AB}$ $\vec{w}$ ✓

$d = \text{Proj of } \vec{AB} \text{ on } \vec{w}$

$$= \frac{w^T \vec{AB}}{\|w\|}$$

→ Perpendicular distance b/w a point & plane.



$d = \text{Proj of } \vec{AB} \text{ on } \vec{w}$

$$= \frac{\vec{w}^T \vec{AB}}{\|\vec{w}\|}$$

$$= \frac{\vec{w}^T (\vec{x}_2 - \vec{x}_1)}{\|\vec{w}\|}$$

$$= \frac{\vec{w}^T \vec{x}_2 - \vec{w}^T \vec{x}_1}{\|\vec{w}\|}$$

$$= \frac{(\vec{w}^T \vec{x}_2 + w_0) - (\vec{w}^T \vec{x}_1 + w_0)}{\|\vec{w}\|}$$

As $\vec{x}_1$ lies on the line $\vec{w}^T \vec{x} + w_0 = 0$

$$d = \left| \frac{\vec{w}^T \vec{x}_2 + w_0}{\|\vec{w}\|} \right|$$

$$\frac{\vec{w}^T \vec{x}_2 + w_0}{\|\vec{w}\|}$$

→ Break until 22:17 PM

→ Perpendicular distance of point x''

from plane $w^T x + w_0 = 0$

↳ $\left| \frac{\vec{w}^T \vec{x}'' + w_0}{\|\vec{w}\|} \right| \rightarrow \text{SVM}$

→ Perpendicular distance of Origin $(0,0)$

from plane $w^T x + w_0 = 0$

→ Perpendicular distance of Origin $(0,0)$

from plane $w^T x + w_0 = 0$

$1/||w||$

$$\left| \frac{w^T x'' + w_0}{||w||} \right|$$

$$= \left| \frac{0 + w_0}{||w||} \right|$$

$$= \left| \frac{w_0}{||w||} \right|$$

Scalar

Scalar (Bias)

$$x'' = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$w^T x'' = 0$$

Vector

→ What is dist. from ~~origin~~ ✓ to
line $3x + 4y - 12 = 0$

1) $7/5$

2) $3/5$

3) $12/5$

4) $4/5$

$$\left| \frac{w_0}{\|w\|} \right|$$

$$= \left| \frac{-12}{\sqrt{3^2 + 4^2}} \right|$$

$$= \frac{12}{5}$$

$$= \frac{12}{5} = \frac{12}{5}$$

$$3x + 4y - 12 = 0$$

$$\left| \frac{3 \times 2 + 4 \times 3 - 12}{\sqrt{3^2 + 4^2}} \right|$$

$$= \frac{6}{5}$$

→ What is $\perp$ dist b/w $(2, -3)$

and line $3x - 4y = 5$

$$\hookrightarrow 3x - 4y - 5 = 0$$

$$w = \begin{bmatrix} 3 \\ -4 \end{bmatrix} \quad w_0 = -5$$

$$x'' = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$w^T x'' = 6 + 12 = 18$$

$$w^T x'' + w_0 = 18 - 5 = 13$$

$$\left| \frac{w^T x'' + w_0}{\|w\|} \right| = \left| \frac{13}{5} \right|$$

$$= \frac{13}{5}$$

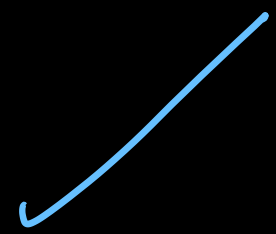
$$\|w\| = \sqrt{3^2 + (-4)^2}$$

$$= 5$$

$$\omega_1 x_1 + \omega_2 x_2 + \omega_0 = 0 \rightarrow \text{Line}$$

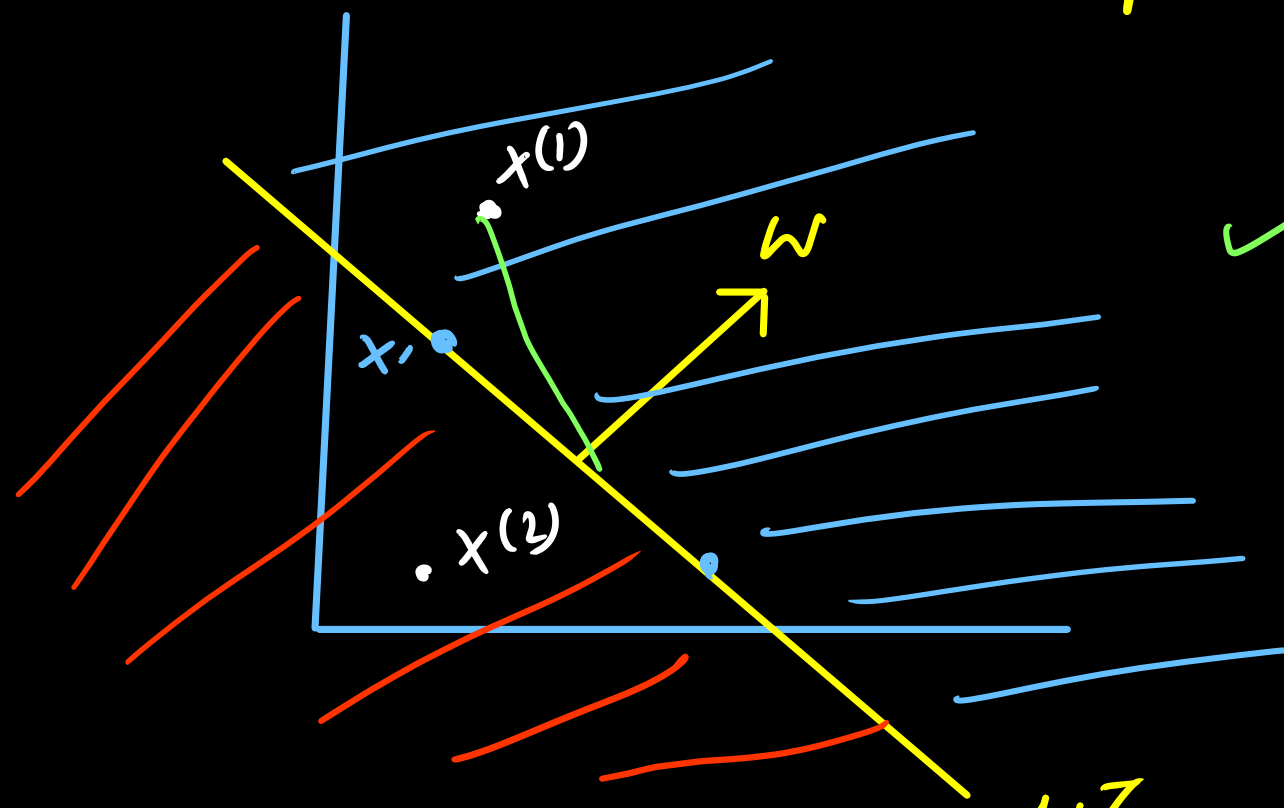
$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \omega_0 = 0 \rightarrow \text{Plane}$$

$$\vec{\omega} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$


$$\omega^T x + \omega_0 = 0$$

→ Half Space

Satisfy $\rightarrow w^T x' + w_0 = 0$



✓ $w^T x^{(1)} + w_0 > 0 \rightarrow +ve$
half space

✓ $w^T x^{(2)} + w_0 < 0$

$w^T x + w_0 = 0$

$w^T \rightarrow -ve$
half space

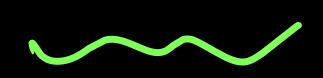
$$w^T x + w_0 > 0 \rightarrow +ve \text{ half}$$

$$< 0 \rightarrow -ve \text{ half}$$

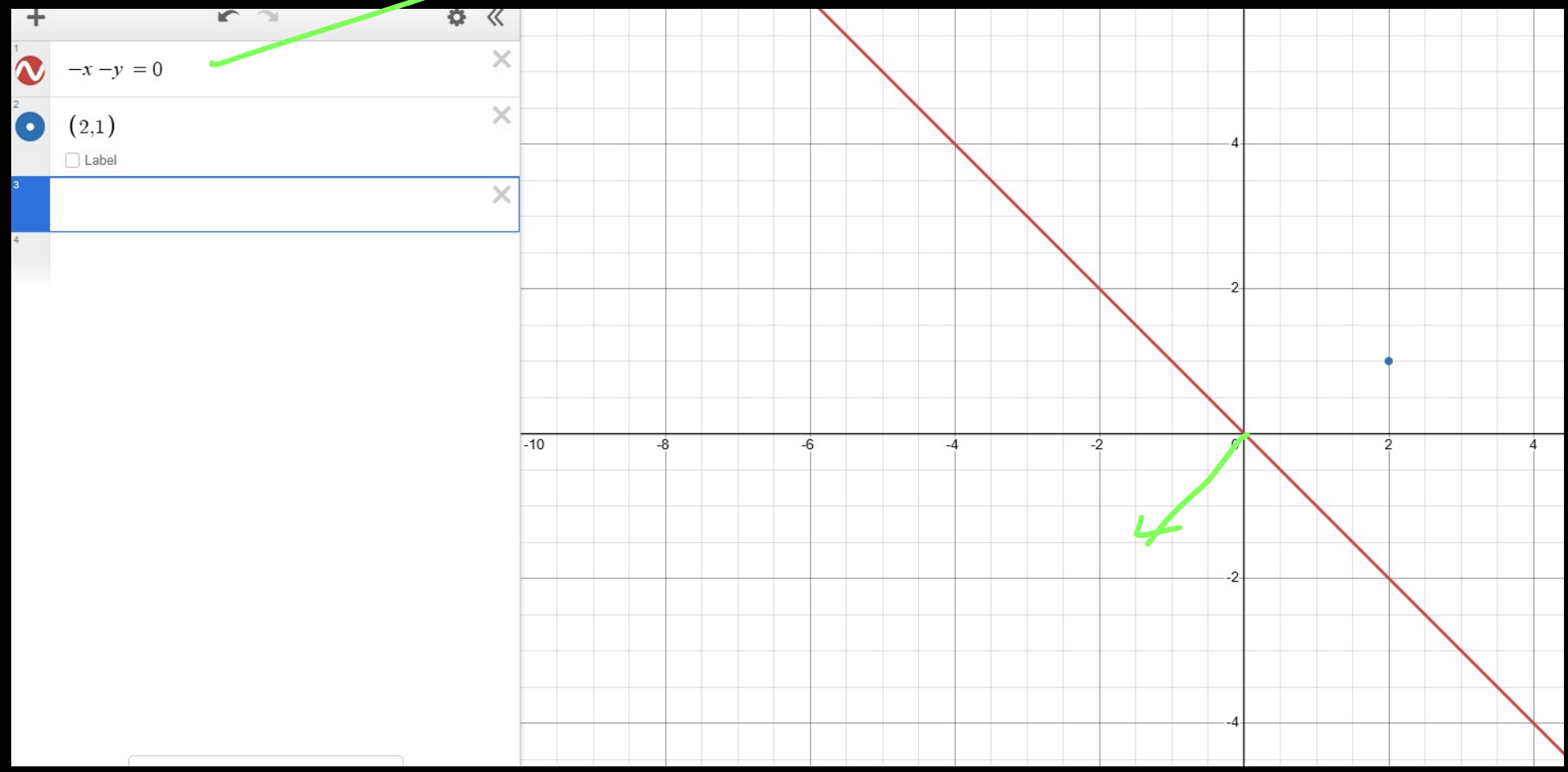
$$-x - y = 0$$

$$w = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$-2 - 1 = -3 < 0$$



-ve
half space



$$x_1 - x_2 = 0$$

$$1 - 4 = -3 < 0$$

